TSA3 Test Report

Executed on 2026-11-11 (Asia/Manila client date) using Eclipse Temurin JDK 25.0.5. All Java sources were compiled with --release 17 -Xlint:all -Werror; no compiler warnings/errors occurred. Assertions execute regardless of -ea. The simplified build script deletes compiled classes before reusing build/classes and build/legacy, so stale classes cannot affect results.

Observed results

- TSA3 service suite: 26 passed, 0 failed.
- Packaged polymorphism regression: 3 passed, 0 failed.
- Regression harness against unchanged TSA2 classes: 4 passed, 0 failed.
- Total assertion-based scenarios: 33 passed, 0 failed.
- Original TSA2 PolymorphismTest: executed successfully; print-only, not counted as additional assertion tests.
- Console demonstration: executed successfully with a fixed date and default UUID identifiers (IDs vary per run).
- SHA-256 preservation checks: all 12 original TSA1/TSA2 Java files matched.

Reproduction and evidence

Run sh build.sh from the TSA3 project root, or follow the individual commands in README. Current execution output is verification-final.txt; installed tool versions are environment.txt. The earlier verification.txt is preserved unchanged as historical evidence. Simplification checks verify all 26 scenario bodies/assertions were retained, all original evidence hashes still match, and TSA1/TSA2 sources are unchanged.

Service assertions are in LibraryServiceTest.java. Each of the 26 scenarios now has a named test method; main only selects/runs scenarios and prints the final count. The extracted method bodies preserve every original setup step and assertion. Each row maps to the exact PASS Txx marker in the captured output. “PASS” records actual execution, not predicted behavior. Every service scenario also runs the consistent helper to check active/history/book/set agreement and reservation identity validity.

T01 - Register members and books

Input / setup: Three member subtypes; four AVAILABLE books; whitespace lookup

Expected result (asserted): 3 members, 4 books; normalized lookup returns original identity

Actual observed result: All listed assertions completed; PASS T01 registration and book addition

Status / evidence: PASS; execution log

T02 - Reject duplicate identities

Input / setup: Re-register S and A with surrounding whitespace

Expected result (asserted): DuplicateEntityException twice; original identity and summary unchanged

Actual observed result: All listed assertions completed; PASS T02 duplicate member and ISBN rejection

Status / evidence: PASS; execution log

T03 - Issue a dated boundary loan

Input / setup: Student borrows A for 14 days on Nov 1

Expected result (asserted): ACTIVE; due Nov 15; loan lookup succeeds; one checked-out ISBN

Actual observed result: All listed assertions completed; PASS T03 successful issuance and due date

Status / evidence: PASS; execution log

T04 - Reject unavailable book

Input / setup: Student holds A; Faculty requests A

Expected result (asserted): ItemUnavailableException; no additional loan or report change

Actual observed result: All listed assertions completed; PASS T04 unavailable book rejection

Status / evidence: PASS; execution log

T05 - Enforce all membership durations

Input / setup: Student 15, Faculty 31, Guest 8 days

Expected result (asserted): LoanLimitExceededException for each; no history entries; all 4 books available

Actual observed result: All listed assertions completed; PASS T05 membership duration violation

Status / evidence: PASS; execution log

T06 - Return and freeze fine

Input / setup: Student 1-day loan on Nov 1; return Nov 5; repeat return

Expected result (asserted): RETURNED; PHP 15 fixed fine; book available; repeat return rejected

Actual observed result: All listed assertions completed; PASS T06 successful and duplicate return with final fine

Status / evidence: PASS; execution log

T07 - Handle unfulfillable queues

Input / setup: Empty queue, then one reservation on checked-out A

Expected result (asserted): Optional.empty in both cases; unavailable reservation stays queued

Actual observed result: All listed assertions completed; PASS T07 empty and wholly unavailable queues

Status / evidence: PASS; execution log

T08 - Handle missing entities

Input / setup: Unknown member, ISBN, loan; issue/return/reserve using missing entities

Expected result (asserted): Lookup returns empty; operations throw NoSuchElementException without mutations

Actual observed result: All listed assertions completed; PASS T08 missing searches and operation entities

Status / evidence: PASS; execution log

T09 - Sort by three keys including ties

Input / setup: IDs 30/20/10/40; returned Nov 2 due date; two overdue Nov 3 ties; one due today Nov 5

Expected result (asserted): Order L-10, L-20, returned L-30, L-40; due today is not overdue; history unchanged

Actual observed result: All listed assertions completed; PASS T09 overdue due-date and ID sorting ties

Status / evidence: PASS; execution log

T10 - Compute aggregates and money

Input / setup: S/F/G loans for 1/2/3 days from Nov 1; report Nov 5

Expected result (asserted): 1 available, 3 checked out, one per type, PHP 29; after S return active fine PHP 14

Actual observed result: All listed assertions completed; PASS T10 aggregate counts groups and fine totals

Status / evidence: PASS; execution log

T11 - Preserve per-book FIFO without global blocking

Input / setup: Queue F:A, G:A, G:B; return B first, then A

Expected result (asserted): G receives B first; A queue unchanged; F then G receive A; direct bypass rejected

Actual observed result: All listed assertions completed; PASS T11 per-book FIFO and independent book progress

Status / evidence: PASS; execution log

T12 - Preserve every collection after failure

Input / setup: Snapshot all six required collections; invalid zero-day and over-limit requests

Expected result (asserted): Both rejected; all six snapshots equal current values; invariants hold

Actual observed result: All listed assertions completed; PASS T12 failed issue leaves every collection unchanged

Status / evidence: PASS; execution log

T13 - Protect collections and domain values

Input / setup: Clear attempts on lists/set/maps; hold old values across issue/return

Expected result (asserted): UnsupportedOperationException for all attempted mutations; old snapshots unchanged

Actual observed result: All listed assertions completed; PASS T13 immutable collection and domain snapshots

Status / evidence: PASS; execution log

T14 - Search and filter accurately

Input / setup: Mixed-case Java keyword; one checked-out book; Guest filter

Expected result (asserted): 2 title matches, 1 checked-out and 3 available; Guest G; result lists immutable

Actual observed result: All listed assertions completed; PASS T14 title searches and status membership filters

Status / evidence: PASS; execution log

T15 - Cancel one entry without reordering

Input / setup: Three entries; cancel middle; repeat cancellation

Expected result (asserted): First and last remain in order; cancelled value returned; missing cancellation rejected

Actual observed result: All listed assertions completed; PASS T15 reservation cancellation preserves remaining order

Status / evidence: PASS; execution log

T16 - Reject invalid reservations

Input / setup: Available unqueued book, own-held book, duplicate, excessive/negative duration

Expected result (asserted): Appropriate exceptions; only original valid reservation remains

Actual observed result: All listed assertions completed; PASS T16 duplicate and invalid reservations

Status / evidence: PASS; execution log

T17 - Correct classification precedence

Input / setup: Valid available, valid unavailable, over-limit unavailable, zero days

Expected result (asserted): APPROVED, RESERVED, DENIED; zero rejected; preview leaves summary unchanged

Actual observed result: All listed assertions completed; PASS T17 classification denial precedence and no mutation

Status / evidence: PASS; execution log

T18 - Preserve queue after processing ID conflict

Input / setup: Constant ID supplier; return L-same while R-same waits

Expected result (asserted): Reused loan ID rejected; queue, books, history, and summary unchanged

Actual observed result: All listed assertions completed; PASS T18 duplicate generated ID leaves queue and books intact

Status / evidence: PASS; execution log

T19 - Reject impossible return dates atomically

Input / setup: Issue Nov 1; move clock back to Oct 31; return

Expected result (asserted): IllegalArgumentException; original active/history values preserved

Actual observed result: All listed assertions completed; PASS T19 return with backward clock is atomic

Status / evidence: PASS; execution log

T20 - Reject invalid input safely

Input / setup: Blank/null inputs, null enum/object parameters, imported CHECKED_OUT book

Expected result (asserted): Argument/null exceptions as specified; book count remains 4; state consistent

Actual observed result: All listed assertions completed; PASS T20 invalid inputs and imported checkout rejection

Status / evidence: PASS; execution log

T21 - Avoid invented count cap

Input / setup: Guest borrows all four different books for 7 days

Expected result (asserted): 4 simultaneous valid loans accepted; checked-out set and history consistent

Actual observed result: All listed assertions completed; PASS T21 no invented simultaneous loan count limit

Status / evidence: PASS; execution log

T22 - Propagate external creation failures

Input / setup: ID supplier throws IllegalStateException before issuance

Expected result (asserted): Same exception type propagates; circulation summary and invariants unchanged

Actual observed result: All listed assertions completed; PASS T22 supplier failure is propagated without state change

Status / evidence: PASS; execution log

T23 - Prevent cancelled ID reuse

Input / setup: Constant ID source; cancel R-same then reserve again

Expected result (asserted): DuplicateEntityException; cancelled ID not reused; queue remains empty

Actual observed result: All listed assertions completed; PASS T23 cancelled reservation IDs cannot be reused

Status / evidence: PASS; execution log

T24 - Report empty state

Input / setup: New service with no entities or circulation

Expected result (asserted): Zero counts/fines; empty sorted list; all three membership groups equal zero

Actual observed result: All listed assertions completed; PASS T24 empty reports include zero membership groups

Status / evidence: PASS; execution log

T25 - Reject stale return after reissue

Input / setup: Return old Student loan; Guest reborrows same ISBN; return old loan again

Expected result (asserted): NoSuchElementException; new loan stays active; history retains 2 records

Actual observed result: All listed assertions completed; PASS T25 stale return cannot release a reissued book

Status / evidence: PASS; execution log

T26 - Retain fairness for available queued books

Input / setup: F waits for A; A returns; G joins queue; cancel F

Expected result (asserted): Queue accepts G behind F; preview RESERVED; after cancellation G is served

Actual observed result: All listed assertions completed; PASS T26 available reserved book retains FIFO after cancellation

Status / evidence: PASS; execution log

Policy regression evidence

PolymorphismTest.java adapts the original Member-reference test with executable assertions.

POLY-1 - Student through Member reference; boundary and unavailable requests; 3 late days

Expected: 14 days; PHP 5/day; PHP 15 fine; exact boundary approved; above denied; unavailable reserved

Actual observed: All policy/outcome/fine assertions passed

Status: PASS

POLY-2 - Faculty through Member reference; same cases

Expected: 30 days; PHP 2/day; PHP 6 fine; correct outcomes

Actual observed: All policy/outcome/fine assertions passed

Status: PASS

POLY-3 - Guest through Member reference; same cases

Expected: 7 days; PHP 10/day; PHP 30 fine; correct outcomes

Actual observed: All policy/outcome/fine assertions passed

Status: PASS

LegacyRegressionTest.java is compiled separately against the untouched TSA2 files.

LEGACY-1 - Original StudentMember, available/unavailable books, 3 late days, return

Expected: 14, PHP 5; APPROVED/RESERVED/DENIED; PHP 15; returned book available

Actual observed: All original policy/classification/fine/return assertions passed

Status: PASS

LEGACY-2 - Original FacultyMember, same cases

Expected: 30, PHP 2; correct outcomes; PHP 6; return available

Actual observed: All assertions passed

Status: PASS

LEGACY-3 - Original GuestMember, same cases

Expected: 7, PHP 10; correct outcomes; PHP 30; return available

Actual observed: All assertions passed

Status: PASS

LEGACY-4 - Original Student 15-day unavailable request

Expected: RESERVED in original TSA2, capturing its precedence defect

Actual observed: RESERVED observed; intentional TSA3 DENIED correction verified by T17

Status: PASS

Original print-only test output is retained in the same execution log (14/30/7 and PHP 5/2/10). No additional pre-existing assertion-based regression suite was present.

Demonstration observations

The console demo approved a 14-day Student boundary request, issued a UUID-identified loan due 2026-11-25, produced a RESERVED preview and UUID-identified reservation, safely handled an unfulfillable queue, rejected Guest 8 days, returned the first loan, and fulfilled the Faculty reservation as a new UUID-identified loan. Final counts were 1 available, 1 checked out, 1 active, 2 historical, 0 pending, and PHP 0.00 active estimated fines.

Exception evidence

Six supplementary captures executed against the same clean-build service classes and recorded the actual exception types/messages below. Each capture also compared all six collection snapshots before and after rejection. These captures illustrate the existing tests; the main suite count remains 33. The capture source and reproduction command are retained in evidence/exception-evidence.txt.

```
E01 / T02 duplicate member
  Actual: DuplicateEntityException: Member ID already registered: S
  Expected type matched; all six collection snapshots unchanged: true
E02 / T02 duplicate ISBN
  Actual: DuplicateEntityException: ISBN already registered: A
  Expected type matched; all six collection snapshots unchanged: true
E03 / T05 duration limit
  Actual: LoanLimitExceededException: Requested 15 days; STUDENT allows at most 14.
  Expected type matched; all six collection snapshots unchanged: true
E04 / T04 unavailable book
  Actual: ItemUnavailableException: Book is already checked out: A
  Expected type matched; all six collection snapshots unchanged: true
E05 / T16 duplicate reservation
  Actual: DuplicateEntityException: Member already has a pending reservation for ISBN A
  Expected type matched; all six collection snapshots unchanged: true
E06 / T08 invalid return
  Actual: NoSuchElementException: Active loan not found: missing
  Expected type matched; all six collection snapshots unchanged: true
Exception captures: 6 verified; all collection snapshots unchanged after rejection.
```

AI assistance disclosure

OpenAI Codex assisted with implementation, review, test execution, and submission documentation. The student should review and understand the submitted code. No manual user verification or instructor acceptance is claimed. See AI_ASSISTANCE_DISCLOSURE.md for the full disclosure.

Verification limits

No screenshots or manual user acceptance tests were produced. The final PDF exports and architecture diagram were rendered and reviewed for readable layout; the SVG source also passed XML validation. Java 17 output compatibility was enforced by the compiler, but tests ran on JDK 25 rather than a separate Java 17 runtime. No instructor PDF was available for comparison. Concurrency, persistence, GUI integration, and runtime-failure recovery are outside TSA3 scope.